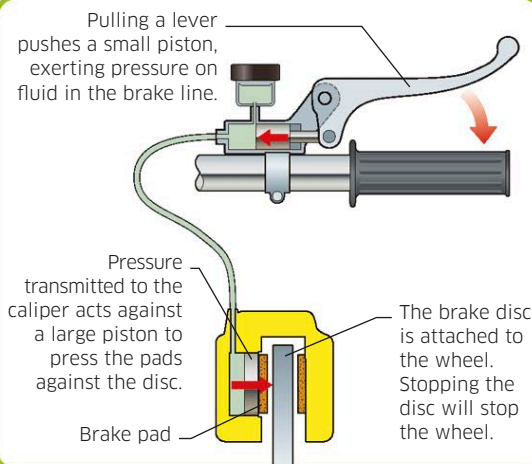


Friction in a motorcycle

The force of friction both helps and hinders a motorcycle rider. Friction between the tires and ground is essential for movement and grip, and is the force behind braking. Drag, the friction that occurs between air and the bike, slows the rider down, and friction between moving parts makes the bike less efficient.

Front fairings

The front of the bike is streamlined so that air flows around it, reducing drag.



How disc brakes work

Most modern motorcycles have disc brakes on their wheels. When the brake lever is pulled, hydraulic pressure (see p.106) multiplies the force to press the brake pads against the disc. Friction between the pads and disc slows or stops the bike, generating heat as “lost” energy.

Hydraulic brake line

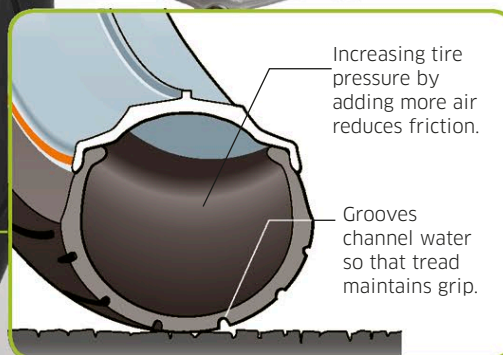
Brake calipers

Brake pads

Most pads are made of metals fused under heat and pressure to create heat-resistant compounds.

Brake discs

Drilled discs help heat produced by friction to escape.

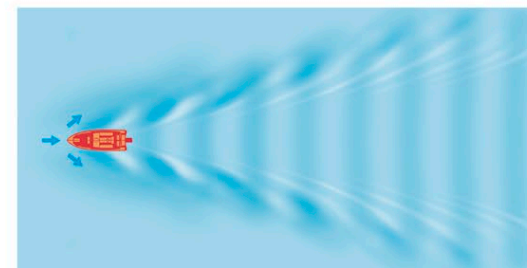


How tread maintains friction

Friction helps the tires to grip the ground as the bike moves, preventing it from skidding. The tread is designed to channel water through grooves, so that the tires still grip on wet and muddy roads.

Fluid resistance (drag)

When an object moves through a fluid, it pushes the fluid aside. That requires energy, so the object slows down—or has to be pushed harder; this is known as form drag. Fluid also creates friction as it flows past the object's surface; this is called skin friction.



Water resistance

When a boat moves through water, it pushes water out of the way. The water resists, rising up as bow and stern waves and creating transverse waves in the boat's wake.

Air resistance

When an object moves through air, the drag is called air resistance. The bigger and less streamlined the object and the faster the object is moving, the greater the drag. When spacecraft re-enter the atmosphere, moving very fast, the drag heats their surfaces to as much as 2,750°F (1,500°C).



Helpful and unhelpful friction

It is tempting to think of friction as an unhelpful force that slows movement, but friction can be helpful, too. Without friction between surfaces, there would be no grip and it would be impossible to walk, run, or cycle. However, the boot is on the other foot for skiers, snowboarders, and skaters, who minimize friction to slide.



Reducing friction

The steel blades of ice skates reduce friction, enabling skaters to glide across ice.



Increasing friction

The treads of rubber-soled mountain boots increase friction and grip for climbers.